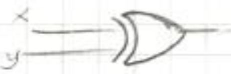
EXOR = XOR Gate

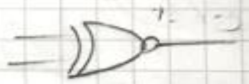
Dec	x.y	$f = x \oplus y$
0	0 0	0
1	0 1	1
2	1 0	1
3	1 1	0

$f = x.\bar{y} + \bar{x}y = x \oplus y$



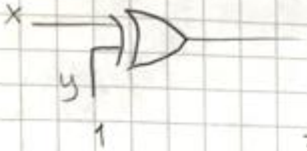
Equivalence = ExNOR = EQV

	x y	$f = x \odot y$
0	0 0	1
1	0 1	0
2	1 0	0
3	1 1	1

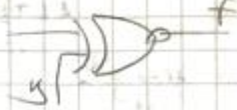


$f = x \odot y = x.y + \bar{x}.\bar{y}$

Controlled Logic level conversion.



$f = x \oplus y = x.\bar{y} + \bar{x}y$



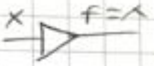
$f = x.y + \bar{x}.\bar{y}$

1 $= \bar{x}$ Conversion Mode ← 0 $f = \bar{x}$

0 x Transfer Mode ← 1 $f = x$

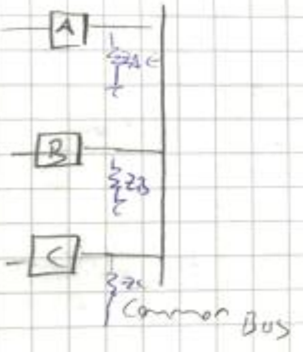
Buffer

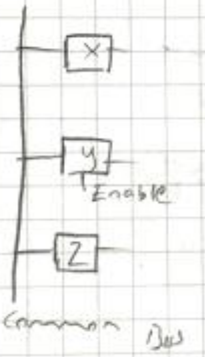
x	y
0	0
1	1



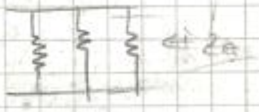
→ Sinyallerin seviyesini değiştirmede ideal hale getiriy.

Tristate Gates

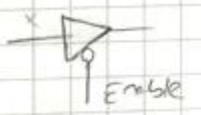

A
B
C
Z_A
Z_B
Z_C
Common Bus

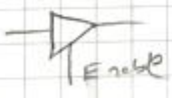

X
Y
Z
Enable
Common Bus

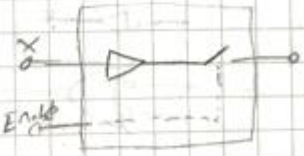
$\frac{1}{Z_{eş}} = \frac{1}{Z_A} + \frac{1}{Z_B} + \frac{1}{Z}$



Tristate Buffer







$\bar{E}$	F
0	X
1	Hi-Z

Hi-Z ↓ Yüksek empedans

E	F
0	Hi-Z
1	X

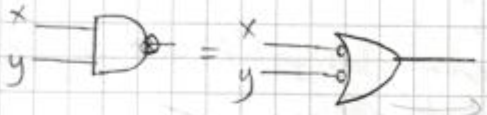
CONJUGATE GATE SYMBOL
(ALTERNATE GATE SYMBOL)

1. operatör değiştiriliy
2. Giriş ve çıkışlar da tümleyenleri alınıyo

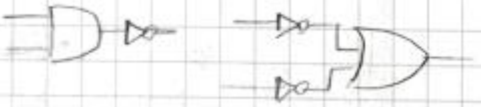
De Morgan

$(\overline{x \cdot y}) = (\bar{x} + \bar{y})$

$(x \cdot y)(L) = (\bar{x} + \bar{y})(H)$



Bu şunun alternate Gate symbolüdür

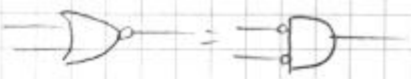


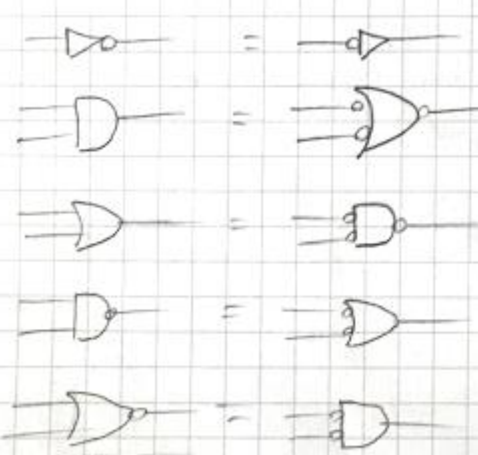
Standard Gate Symbol

⇓

Alternate Gate symbol

$\overline{x + y} = \bar{x} \cdot \bar{y}$

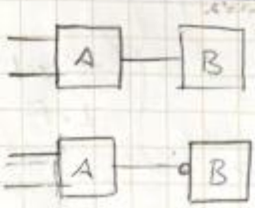


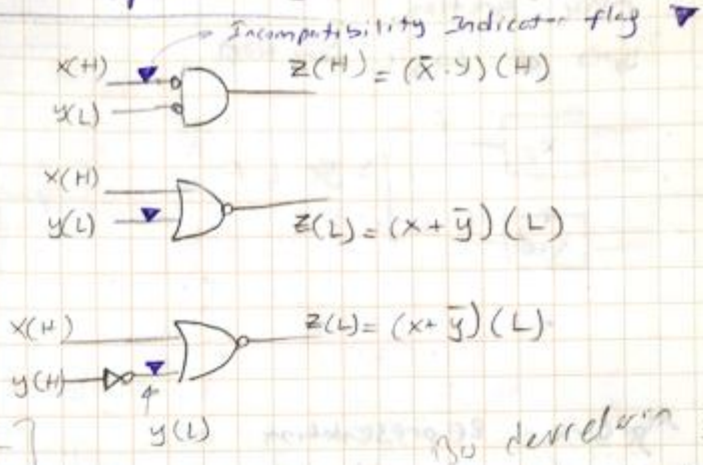


Standard

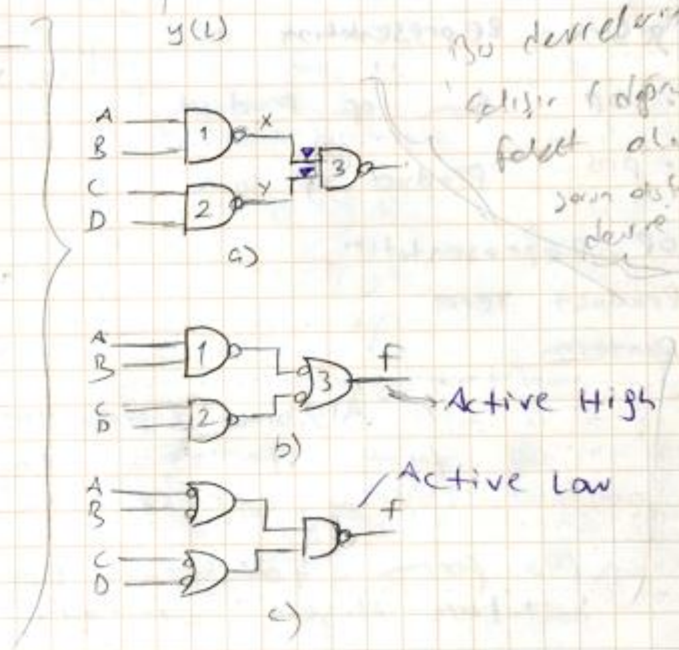
Alternate.

Logic Level Incompatibility (Lojik seviye uyumsuzluğu)


Seri Circuit
A
B
A
B


Incompatibility Indicator flag ▼
X(H)
Y(L)
Z(H) = (X̄ · Y)(H)
X(H)
Y(L)
Z(L) = (X + Ȳ)(L)
X(H)
Y(H)
Y(L)
Z(L) = (X + Ȳ)(L)

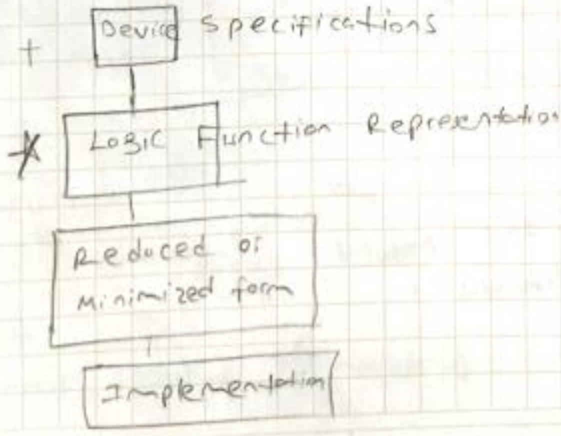
dec	ABCD	f
0	0 0 0 0	0
1	0 0 0 1	0
2	0 0 1 0	0
3	0 0 1 1	1
4	0 1 0 0	0
5	0 1 0 1	0
6	0 1 1 0	0
7	0 1 1 1	1
8	1 0 0 0	0
9	1 0 0 1	0
10	1 0 1 0	0
11	1 0 1 1	1
12	1 1 0 0	1
13	1 1 0 1	1
14	1 1 1 0	1
15	1 1 1 1	1

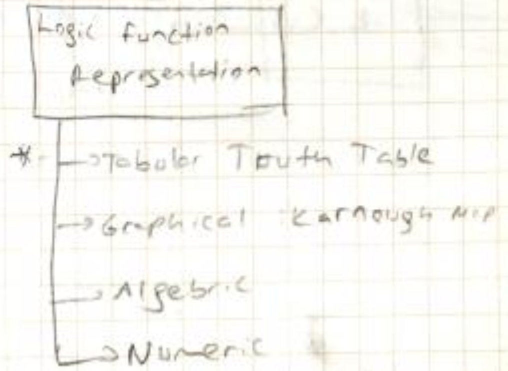

A
B
1
X
C
D
2
Y
3
a)
A
B
C
D
1
2
3
f
Active High
b)
Active Low
A
B
C
D
f
c)

Bu devrelerin } çalışır (doğrudur) fakat alırsa. sorun olsaydı devre olur

Alternative sembolleri kullanarak lojik uyumsuzluğu kaldırıyor.

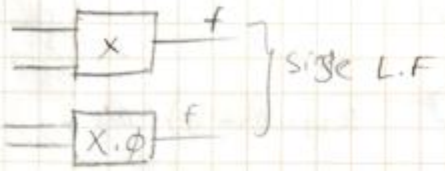
LOGIC FUNCTION

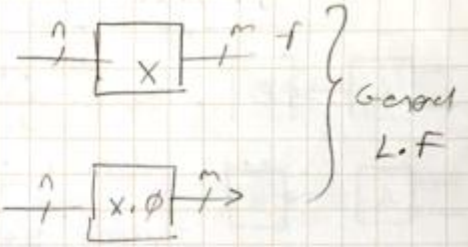

+
Device specifications
*
Logic Function Representation
Reduced or Minimized form
Implementation


Logic function Representation
*
Tabular Truth Table
Graphical Karnaugh Map
Algebric
Numeric

Logic Function

Types of Logic functions





Algebric Representation

- SOP Sum of product
- POS Product of sum

SOP Representation

Product Term

Minterm

Algebric Representation

SOP → SOP form fonksiyonun Logic 1 değerini aldığı noktala ilişkin bir gösterimi,

POS → POS form fonksiyonun Logic 0 değerini aldığı noktalara ilişkin bir gösterim

SOP presentation:

- A, B, C Literal
- AB.AC, BC, $\bar{A}B$, $A\bar{C}$, $\bar{B}\bar{C}$ } Product Term
- ABC, $\bar{A}BC$, $A\bar{B}C$, ... $\bar{A}\bar{B}\bar{C}$ } Product Term

→ Minterm → Girişleri 3'ünü de içerdi isn minterm değil

Connonical product Term

fundemental " "

Minterm = connonical product term = fundemental product term minterm code

Uncomplemented variable x logic 1

complemented variable $\overline{X}$ logic 0

$m_i = \overline{A}BC = m_3$

$(011) \rightarrow 3$

↓

Cannonical SOP form → Eğer çarpım terimlerinin tümü minterm ise buna der..

~~örnek~~

$f(A,BC) = \overline{A}B\overline{C} + AB\overline{C} + \overline{A}BC + ABC$

Cannonical sop form $(010) + (110) + (011) + (111)$

$$\begin{array}{cccc} \downarrow & \downarrow & \downarrow & \downarrow \\ 2 & 6 & 3 & 7 \\ m_2 & m_6 & m_3 & m_7 \end{array}$$

Numerical Representation $= m_2 + m_3 + m_6 + m_7$

$= \sum m(2,3,6,7)$

Logic fonksiyonların nümerik gösterimi

Minterm	DEC	A B C	
m_0	0	0 0 0	0
m_1	1	0 0 1	0
m_2	2	0 1 0	1
m_3	3	0 1 1	1
m_4	4	1 0 0	0
m_5	5	1 0 1	0
m_6	6	1 1 0	1
m_7	7	1 1 1	1

$f = \sum m(2,3,6,7)$

$f = \overline{A}B\overline{C} + \overline{A}BC\ AB\overline{C} + ABC$

Bu üç gösterimden hangisi elimizde ise değerlerine geçiş yapabiliriz

for generating the cannical SOP form

$f(A,B,C) = A + \overline{B}C + \overline{A}BC$ → Cannonical'e çevirme

1) $X + \overline{X} = 1$

$f(A,B,C) = A(B+\overline{B})(C+\overline{C}) + (A+\overline{A})\overline{B}C + \overline{A}BC$

2) Distributive

$f(A,BC) = ABC + AB\bar{C} + A\bar{B}C + A\bar{B}\bar{C} + A\bar{B}C + \bar{A}\bar{B}C + \bar{A}BC$

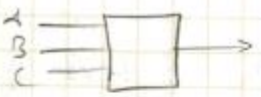
3) $x + x = x$

$f = ABC + AB\bar{C} + A\bar{B}C + A\bar{B}\bar{C} + \bar{A}\bar{B}C + \bar{A}BC$

$f = \sum m(7,6,5,4,1,3)$

$f = \sum m(1,3,4,5,6,7)$

POS (Product of sum) Representation



- $A, B, C, \bar{A}, \ldots \bar{C}$ literal
- $A+B, A+C, B+C, \bar{A}+B \ldots \bar{B}+\bar{C}$
- $A+B+C, A+\bar{B}+\bar{C}, \ldots \bar{A}+\bar{B}+\bar{C}$ } Bunlar Maxterm

Sum Term

Maxterm $M_i \Rightarrow$ Maxterm code

Complemented variable $\bar{x}$ logic 1
Uncomplemented variable x logic 0 } $\bar{A}+\bar{B}+C+\bar{D} = M_{13}$

$\underbrace{1\ 1\ 0\ 1}_{13}$

maxterm dan

$\overline{m_i} = M_i$ $\quad m_5 = A\bar{B}C$

$\overline{M_i} = m_i$ $\quad \overline{m_5} = \overline{A\bar{B}C} = \bar{A}+B+\bar{C} \rightarrow M_5$

Cannonical POS form → Bütün terimleri Maxterm olan

$$f(A,B,C) = \underset{M_1}{\underset{0\,0\,1}{(A+B+\bar{C})}} \cdot \underset{M_5}{\underset{1\,0\,1}{(\bar{A}+B+\bar{C})}} \underset{M_4}{\underset{1\,0\,0}{(\bar{A}+B+C)}} \cdot \underset{M_0}{\underset{0\,0\,0}{(A+B+C)}}$$

$= \prod M(0,1,4,5)$

↓ [illegible]

Dec	A	B	C	f
0	0	0	0	0
1	0	0	1	0
2	0	1	0	1
3	0	1	1	1
4	1	0	0	0
5	1	0	1	0
6	1	1	0	1
7	1	1	1	1

Conversion Between POS and SOP forms

SOP form
POS form

1) from SOP to POS form

$f(x,y,z) = x\bar{z} + \bar{x}y + \bar{y}z$

De Morgan teoremini

$f(x,y,z)=x\bar{z}+\bar{x}y+\bar{y}z \Rightarrow f^d(x,y,z)=(\bar{\bar{x}}+z)\cdot(\bar{x}+\bar{\bar{y}})\cdot(y+\bar{\bar{z}})$

$f^d(x,y,z)=\bar{x}\bar{y}\bar{z}+xyz$

$f^{d^d}(x,y,z)=f(x,y,z)=(\bar{x}+\bar{y}+\bar{\bar{z}})\cdot(\bar{x}+y+z)$

2)- from POS to SOP form

$f(x,y,z)=(x+y)(\bar{x}+z)=x\cdot\bar{x}+x\cdot z+\bar{x}y+yz$

$=xz+\bar{x}y+yz$